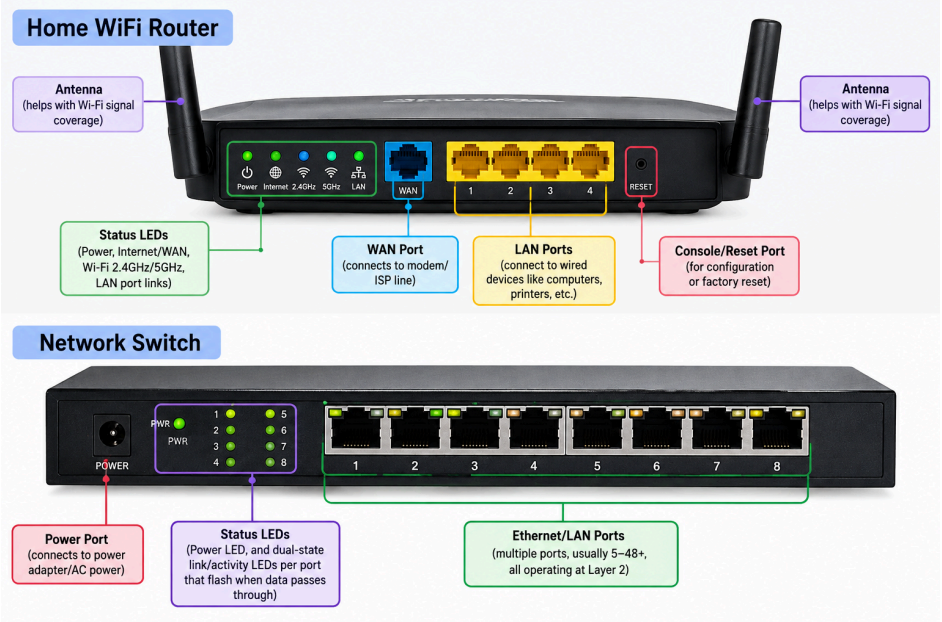
Assignment-4.2

“Router and Switches”

TASK-1 Draw a labelled diagram of a typical home WiFi router and a network switch, clearly marking at least 3 different ports (such as WAN, LAN, Console) and the location of status LEDs.

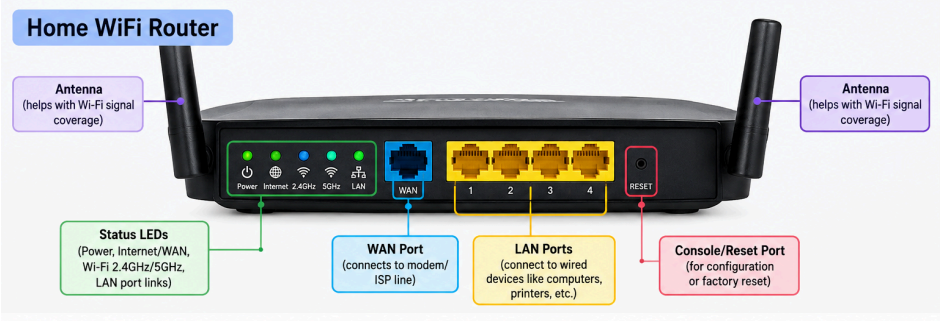


TASK-2 List 3 key differences between routers and switches in terms of their main functions and how they handle network traffic. Think about how each device deals with IP addresses, MAC addresses, and the types of networks they connect.

Basis of Difference	**Router**	**Network Switch**
1. Main Function	Connects different networks, such as a home or office network to the internet. It helps devices communicate with devices on other networks.	Connects multiple devices, such as computers and printers, within the same local area network (LAN). It allows them to communicate and share data.
2. Addresses Used	Uses **IP addresses** to identify the destination network and	Mainly uses **MAC addresses** to identify devices on a local

	decide where data should be sent.	network and forward data to the correct port.
3. Handling Network Traffic	Forwards data between different networks. For example, it sends requests from a computer to an internet server and returns the response.	Forwards data between connected devices within a local network. It learns the MAC addresses of connected devices to send data through the appropriate port.

TASK-3.Take a real or virtual image of a router and a switch (from your home, lab, or Google Images), and identify the following on each: Power port, at least 2 Ethernet ports, and any LED indicators. Annotate the image or describe their position and purpose.



1. Power Port

- **Location:** Usually found on the back or side of the router.
- **Purpose:** The power adapter connects to this port to supply electricity to the router. Without power, the router cannot operate.

2. Ethernet Ports (LAN)

- **Location:** Usually found on the back of the router. They are often grouped together and may be labelled LAN.
- **Purpose:** These ports connect wired devices, such as desktop computers, printers, and network switches, to the local network using Ethernet cables.
- **Example:** An Ethernet cable can connect a desktop computer directly to the router for internet access.

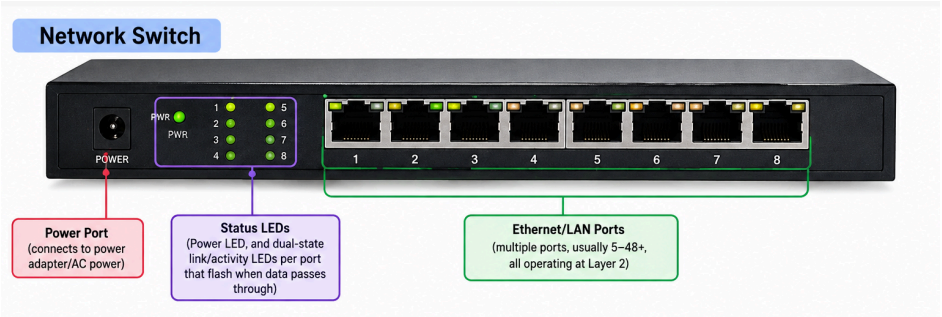
3. WAN Port (Internet Port)

- **Location:** Usually found on the back of the router, separate from the LAN ports. It may be labelled WAN or Internet.
- **Purpose:** This port connects the router to a modem or ONT, which provides the internet connection from the Internet Service Provider (ISP).

- **Example:** The internet connection enters the router through the WAN port, allowing connected devices to access online services.

4. LED Indicators

- **Location:** Usually found on the front or top of the router, although some models place them near the ports.
- **Purpose:** These lights show the router's working status. Different lights may indicate power, internet connectivity, Wi-Fi activity, and wired connections.
- **Example:** A power light shows whether the router is switched on, while an internet light helps indicate the status of its internet connection.



A network switch connects multiple wired devices within a local network. It allows computers and other connected devices to exchange data efficiently.

Parts to Identify:

1. Power Port

- **Location:** Usually found on the back or side of the switch, depending on its design.
- **Purpose:** It connects to a power adapter or power supply to provide electricity to the switch.
- **Example:** When the switch is powered on, it can begin connecting the devices attached to its Ethernet ports.

2. Ethernet Ports

- **Location:** Commonly found in a row on the front of the switch. The number of ports depends on the model.
- **Purpose:** These ports connect computers, printers, servers, routers, and other wired network devices using Ethernet cables.
- **Example:** In a computer lab, eight computers can connect to an eight-port switch, provided all eight ports are available for those computers.

3. LED Indicators

- **Location:** Usually located above or below each Ethernet port, or beside the ports.
- **Purpose:** These lights indicate whether a device is connected and may show network activity. Their exact meaning depends on the switch model.

- **Example:** If a computer is connected and communicating through a port, its LED may light up or blink to show activity.

TASK-4.Imagine you are setting up a small gaming café network for IPL streaming and multiplayer matches. Write a short scenario explaining which device (router or switch) you would connect first to the ISP line, and how you would connect 8 gaming PCs to the network. Justify your choices.

Step 1: Connect the ISP Line

First, I would connect the internet service provider's line to the modem or ONT, if required by the ISP. Then, I would connect the modem or ONT to the router's WAN port using an Ethernet cable.

Step 2: Connect the Network Switch

Next, I would connect one of the router's LAN ports to an Ethernet port on the network switch. This would allow the switch to connect all eight gaming PCs to the network.

Step 3: Connect the Eight Gaming PCs

I would connect each gaming PC to a separate Ethernet port on the switch using an Ethernet cable. An 8-port switch can connect eight PCs, but it would need an additional port for the router connection. Therefore, I would choose a switch with at least 9 available ports, such as a 16-port switch, to connect all eight PCs and the router.

Step 4: Test the Network

Finally, I would turn on the devices and check that all eight PCs have network access. I would test the internet connection, IPL streaming, and multiplayer games.

Why I Chose These Devices

- **Router:** It connects the gaming café's local network to the internet and directs data between networks.
- **Switch:** It connects all eight gaming PCs within the local network and allows them to communicate.
- **Ethernet Cables:** They provide stable, reliable wired connections, which are useful for online gaming and streaming.

Conclusion: I would connect the ISP line to the modem or ONT, then connect the router, followed by the network switch and eight gaming PCs. This setup provides internet access and reliable wired connections for gaming and IPL streaming.